

4 (a) For a body moving in uniform circular motion, there is a change in velocity as the direction of motion is changing.

B1

B1

The rate of change of velocity gives rise to an acceleration. Since the change in velocity is directed towards the centre of the circular motion, the acceleration is also towards the centre.

Comments:

This part is generally quite badly done by most students. Though they are able to explain why there is an acceleration, they were not able to explain why the acceleration is towards the centre.

(b) (i)
$$a = r\omega^2 = (5.0 \cos 60^\circ) \left(\frac{2\pi}{3.5}\right)^2$$
$$= 8.06 \text{ m s}^{-2}$$

M1
A0

Comments:

This part is generally very well done by most students.

(ii)
$$T \cos 60^\circ = ma$$
$$T = \frac{0.40 \times 8.06}{\cos 60^\circ} = \frac{0.40 \times 8.06}{0.5} = 6.45 \text{ N}$$

B1
A1

Comments:

Some of the students resolve the tension T wrongly, hence failing to obtain the marks.

(iii)
$$L = mg + T \sin 60^\circ$$
$$= (0.40 \times 9.81) + 6.45 \sin 60^\circ$$
$$= 9.51 \text{ N}$$

B1
A1

Comments:

Quite a number of students balance the vertical forces wrongly.

(iv) The tension in the wire is unchanged, since centripetal force is unchanged.

M1

The vertical component of tension together with the new lift force now balance the weight of the toy plane hence new lift force will be lesser than before.

A1

Comments:

Many students did not mention the anchor point that tension remains unchanged. Note: Students are not required to know the origin of lift force or how it is related to velocity of air (i.e. Bernoulli effect).